BITPREDICT - A DIGITAL CURRENCY INSIGHT PROGRAM

A Research Study

Presented to the Faculty of the

College of Information and Computer Studies

De La Salle University - Dasmarinas

City of Dasmarinas, Cavite

In Partial Fulfillment of the Requirements for

S-CSPC212 Discrete Structures 1

Gartner, Markus Ivan P.

Talatala, Jeremie Leo J.

Ubungen, John Daniel L.

December 2025

ABSTRACT

The rapid growth of the cryptocurrency market has introduced significant volatility and unpredictability; with this, investors lack knowledge of analytical tools to navigate this landscape. Several studies have shifted from static historical charts to more advanced ones that use real-time data and predictive modeling. However, existing software solutions prioritize complex algorithms and interface design over mathematical logic, and while other studies have delved into solutions involving discrete concepts, their studies are more theoretically focused. As such, this study bridges that gap by introducing BitPredict, a digital currency insight program that applies discrete structures, mainly Graph Theory, Propositional Logic, Set Theory, Algorithms, and Combinatorics, to provide investors with mathematically derived market insights. The prototype was developed using Python, utilizing libraries such as NetworkX for correlation Graphs and Pandas for data handling, with Streamlit serving as the interactive user interface. Real-time data for major assets (BTC, ETH, XRP) was sourced using CCXT API to connect to the Binance US exchange. Based on testing, the three cryptocurrencies yielded correlation rates of 0.89 - 0.97, indicating limited diversification benefits during the observed period of time. Furthermore, the prototype successfully identified a bearish trend through a "death cross" pattern that correctly advised users not to invest during an unstable market. These findings prove that BitPredict effectively visualises complex market behaviors and helps users make responsible decisions.

Keywords: Cryptocurrency, Discrete Structures, Python, Streamlit, Market Analysis

TABLE OF CONTENTS

CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

Introduction

In recent years, the use of cryptocurrency as a new form of digital currency has paved the way to reshape the current landscape of finance and investing. The rapid growth of different cryptocurrencies, such as Bitcoin, Ethereum, and more, gained significant attention from investors. However, the unpredictable nature of cryptocurrency makes it an additional obstacle for investors. These challenges require investors to have sufficient knowledge when it comes to utilizing analytical tools to make proper decisions. The high-risk, although high-reward, environment demands investors take a strategic approach to investing that is supported by different insights and analytics.

The project "**BitPredict: Digital Currency Insight Program**" aims to allow users to access tools that will allow them to navigate the cryptocurrency market much more effectively. By utilizing different mathematical and computational concepts such as graph theory, combinatorics, propositional logic, and set theory, BitPredict provides its users with insights into current cryptocurrency trends. The program analyzes the connection between different cryptocurrencies, provides forecasts on possible value trajectories, and suggests an optimal investment plan to enhance portfolio performance. Additionally, it will allow users to be educated on market instability and risks that will help them make responsible investment practices.

Making use of Python as the sole programming language, BitPredict combines core programming libraries including **NetworkX** and **MatplotLib** for the creation of correlation graphs, **NumPy** for statistical computing, **Itertools** for generating possible portfolio combinations, and **Pandas** for efficient data handling. Moreover, the real-time cryptocurrency data is sourced from the **CCXT** (**CryptoCurrency eXchange Trading**) **API**, which will ensure accurate and relevant market information for analysis. **Streamlit** serves as the main framework for the program due to its diverse tools for displaying data effectively and its compatibility with Python.

With this project, users are provided with a reliable platform that will help promote proper decision-making in investments and enhance the understanding of market behaviors and risk management. With the combination of computing methods with practical applications in the financial market, BitPredict contributes to a strategic and responsible approach when investing in digital currency.

Problem Statement

1. **Unpredictability of the cryptocurrency market** - The cryptocurrency market is highly volatile, frequently experiencing sharp upward and downward trends in short periods of time. For instance, just this year, Bitcoin, the biggest cryptocurrency on the market, fell in value by USD30000 in just one month, further highlighting the unpredictability of the cryptocurrency market and the difficulty in predicting its trajectory.
2. **Confusion on portfolio combinations** - Investors, especially beginners, face a significant obstacle in choosing the most effective choices for their portfolios. As of September 2025, there are more than 10,000 active cryptocurrencies being traded, creating an overwhelming environment where users have trouble picking which one. However, even when investors attempt to narrow down their options to a few assets, they may struggle deducing the best possible choices for their portfolios without the right tools to analyze market trends.
3. **Difficulty Identifying Connections Between Cryptocurrencies** - In the crypto market, there are patterns cryptocurrencies follow that are not obvious to the naked eye. The biggest example of this is how several known coins follow Bitcoin's price trends. During a 40-day rolling window in 2023, Bitcoin and Ethereum, the two largest cryptocurrencies, had an 82% correlation to each other, even going as high as 95%. This poses a risk to those who want to diversify their portfolio but cannot discern the pattern. If they choose multiple cryptocurrencies that closely follow each other, they could be left unprotected in the event of a market crash with no alternative options to counter their monetary losses.

Objectives

The project **BitPredict: Digital Currency Insight Program** seeks to accomplish the following:

1. Inform users on the possible value trajectory of the cryptocurrency market and what actions to take accordingly.
2. Visualize the correlation tendencies and patterns of different cryptocurrency options to help users diversify their portfolio.
3. Suggest possible cryptocurrency investment combinations to boost the portfolio value of users.
4. Educate users on market volatility and the potential risks of investing in cryptocurrency to encourage responsible investments among users.

CHAPTER 2

REVIEW OF RELATED LITERATURES

Related Works

The inherently volatile and unpredictable nature of the cryptocurrency market has necessitated the development of robust computational tools capable of analyzing market trends and complex data structures. Initial iterations of these tools primarily focused on standard technical analysis, relying on historical charts and statistical indicators to track price movements. However, as the market has matured, research has shifted toward developing systems that integrate real-time data visualization with predictive modeling. This evolution began by leveraging machine learning for high-level forecasting and structured blockchain analysis, with contemporary solutions increasingly prioritizing web-based accessibility to enable both technical and non-technical users to interpret market dynamics.

In the domain of web-based analysis, Indhumathi and Kumar (2023) developed a cryptocurrency application utilizing React to provide real-time market insights and visualization dashboards. Their system integrates live cryptocurrency APIs and user-centric interfaces designed to simplify investment monitoring. However, while this study demonstrates the efficacy of front-end interactivity and user experience (UX) in making data accessible, it places less emphasis on the explicit mathematical logic underlying the prediction mechanisms.

Expanding into predictive algorithms, Shewarade et al. (2025) introduced CryptoProphet, a predictive portfolio mobile application. Their research utilized deep learning models, specifically Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU), to generate "Individualized Modeled Strategies" (IMS) for 30 different cryptocurrencies. While Shewarade et al. successfully demonstrated the application of advanced deep learning algorithms for portfolio management, their study primarily focuses on model selection and mobile deployment rather than the fundamental discrete structures governing the data relationships.

However, there have been documented cases of studies using discrete concepts for analyzing cryptocurrency trends. Puerto et al. (2021) used combinatorial optimization to filter scenarios for picking portfolio options, although they used other financial datasets as subjects instead of cryptocurrencies. Nguyen et al. (2023) also confirmed that graph theory is commonly used in analyzing the strength of relationships between different currencies and demonstrated its use to uncover patterns between different options that are normally unseen. These studies highlight the potential and practical utility of using discrete concepts in analyzing market price trends, but their studies were theoretical and did not integrate the methodologies they researched into a public-facing software application.

Building upon those methodologies, the **BitPredict** program demonstrates the application of deterministic algorithms and discrete mathematical structures to cryptocurrency analysis. The project utilizes Python libraries such as NetworkX and Itertools to implement graph theory and combinatorial optimization. The system models market relationships as a weighted graph, where nodes represent assets and edges denote the calculated Pearson correlation coefficient of their price movements. Furthermore, BitPredict employs discrete logic to classify market trends, specifically utilizing Moving Average (MA50/MA200) crossovers, and generates portfolio combinations via algorithmic scoring.

Synthesis

The reviewed literature reveals a clear trajectory in cryptocurrency analysis tools, moving from static technical analysis to dynamic, real-time systems driven by web technologies and machine learning.

1. The Trend: Both **Indhumathi and Kumar (2023)** and **Shewarade et al. (2025)** demonstrate that modern tools rely heavily on advanced frameworks, whether it is React for visualization or Deep Learning for prediction, to process market volatility for the user. Additionally, **Puerto et al. (2021)** and **Nguyen et al. (2023)** have proven that Discrete Structures concepts have potential in analyzing trends in the cryptocurrency market through graph theory and combinatorics.
2. The Gap: However, a gap exists in the practical application and accessibility of discrete structures concepts in these studies. Indhumathi and Shewarade's studies

focus more on either the **User Interface** (visuals and ease of use) or complex **Deep Learning models** (advanced but opaque algorithms) instead of the mathematical logic and concepts that connect those data points. While Puerto and Nguyen validated the use of discrete concepts like graphs and combinatorics with graph theory even being commonly used in correlation analysis, their studies do not apply the said concepts into an app that could help investors decide on the best action to take in the context of the current market.

3. Our Approach: **BitPredict** bridges this gap by combining the concepts studied by Nguyen and Puerto with the interactivity and knowledge of apps such as those of Indhumathi and Shewarade. By applying Discrete Structures, the project provides a transparent view of how mathematical concepts explain market trends, distinguishing it from works that focus purely on commercial application or front-end design.

CHAPTER 3
METHODOLOGY

Methodology

The program uses Python as its programming language with some CSS applied for certain design and layout changes. Python is a popular option to use for data analysis due to its ease of use and readability which makes it easier to read and write code, and there is also an abundance of libraries, such as NumPy, Matplotlib, and NetworkX, that could be used in fetching, analyzing, and plotting data in a visually pleasing way. Additionally, the program uses Streamlit as its framework due to its ease of use, compatibility with Python, and abundance of tools effective for displaying data. However, Streamlit imposes certain rigidities on design and layout, requiring the use of columns to correctly align elements or special CSS commands to override Streamlit's layout constraints.

The program applies several discrete structures concepts to analyze cryptocurrency trajectories and suggest investment strategies for users, such as:

1. **Algorithms** - Algorithms are the main tools of the program in selecting cryptocurrencies. In this prototype, the algorithm uses the Simple Moving Average (SMA), specifically the 200-day Moving Average (200MA) and the 50-day Moving Average (50MA), which is a commonly used indicator of market momentum. The SMA of a cryptocurrency is calculated by adding up the daily closing prices over a certain period of time and then dividing the total by the number of days observed (in this case, 200 days for the 200MA and 50 days for the 50MA). The 50MA represents the short-term trend of the cryptocurrency since it is more perceptive to changes in the price due to only covering 50 days. Meanwhile, the 200MA represents the long-term trend and has a more smoothed out trajectory than the 50MA. The program fetches 463 days of OHLCV data, a standard financial structure recording the Open, High, Low, Close, and Volume of an asset for each trading day, to give the

program sufficient time to calculate the 200MA and 50MA, then compares the two values over the previous 90 days.

2. **Graph theory** - In the program, graph theory is used to represent the relationships between cryptocurrencies. The program displays a graph involving all the available cryptocurrency options, where the assets are represented as nodes, while the edges are the relationships between the movement patterns of different assets. Each edge has a weight tied to it representing the strength of each cryptocurrency's trend relationship with the other. The weights utilize the Pearson Correlation Coefficient, a value representing the relationship between two variables that is calculated by dividing the covariance of the two variables by the product of their standard deviations, through NumPy. The highest value is 1, showing that the two connected stocks have a direct relationship (they rise and fall in the exact same manner), while -1 is the lowest value, showing that the two connected stocks have an inverse relationship (if one goes up, the other goes down).
3. **Propositional logic** - The classification of a stock to a certain set relies on the use of propositional logic. This program applies it in the classification of cryptocurrency trends through the 200MA analysis method, which compares 3 parameters: the 200MA, the 50MA, and the closing price of the stock (P). The classifications used are:
    a. **Bull** - a strong upward trend where the short-term trend is strong and exceeds the long-term trend.
        - Logic: 50MA > 200MA and P > 50MA. The point where the 50MA goes above the 200MA is called a "**golden cross**", signifying a bull market.

    b. **Correction** - long-term trend is up, but there is a slight dip. Weaker upward trend than bull.
        - Logic: 50MA > 200MA and P <= 50MA

    c. **Bear** - a strong downward trend where the short-term trend is weak and lower than the long-term trend
        - Logic: 50MA < 200MA and P <= 50MA. The point where the 50MA goes below the 200MA is called a "**death cross**", signifying a bear market.

d. **Bear rally** - the long-term trend is down, bull price experiences an upturn
    - Logic: 50MA < 200MA and P > 50MA
e. **Neutral** - insufficient data or the market is in complete equilibrium with no crossovers
    - Logic: 50MA = 200MA or 200MA is null

4. **Set theory** - The program applies this alongside propositional logic for classification. Each cryptocurrency is sorted into 5 sets based on their movement trends in the last 90 days: bull, correction, neutral, bear rally, and bear. The program then uses a points system for each classification (+2 for bull markets, +1 for corrections, 0 for neutral stocks, -1 for bear rallies, and -2 for bear markets), which is the main system used for choosing the best options for the user's portfolio.
5. **Combinatorics** - In the program, combinatorics is used to calculate the number of different portfolio combinations a user could choose for their investments. It uses Itertools to calculate every possible portfolio combination with the available options and uses the set scoring system to suggest the best portfolio combination based on current price trends.

Prototype Design

The development of BitPredict focuses on translating theoretical discrete structures into a functional, user-centric application. The prototype was constructed using Python 3.11 as the primary backend language due to its extensive support for data science libraries. For the user interface, Streamlit was selected as the framework to ensure real-time interactivity and dynamic data visualization. The design priority was to create a transparent system where the mathematical logic, specifically graph theory correlations and propositional logic trends, is not hidden but visually presented to the user to aid in decision-making.

1. How the System Works

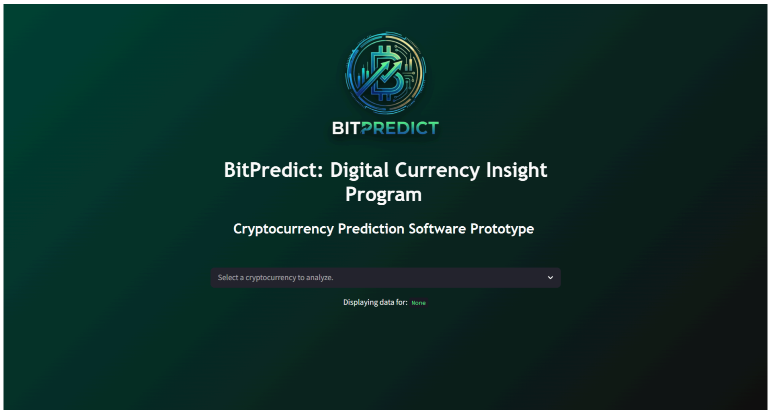

Figure 1. BitPredict Landing Page and User Interface Entry Point.

The application is built on an architecture that separates data fetching, logic processing and front-end presentation.

- Data Layer: The system uses the CCXT library to connect with the Binance US exchange. This allows the program to fetch historical data for major assets. The program collects 463 days of cryptocurrency price data to give the 200MA and 50MA calculations enough time to collect sufficient data, but the program slices the data to display only the last 90 days.
- Processing: Pandas is utilized to organize the raw data into structured tables, while NumPy handles statistical calculations.
- Display: Streamlit renders these calculations into an interactive web interface.

As shown in Figure 1, the landing page provides a streamlined entry point where users can select a specific cryptocurrency asset from a dropdown menu to initiate the analysis. The choice of assets for this prototype is BTC/USDT, ETH/USDT, and XRP/USDT, which are the major market capitalization assets.

2. **Trend Classification Algorithm**

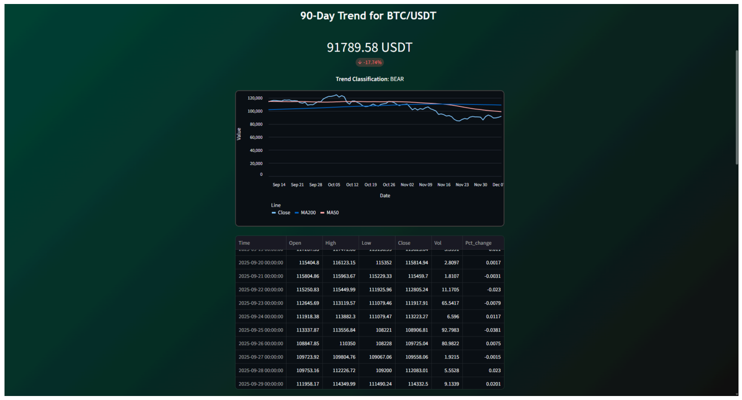

Figure 2. Single cryptocurrency asset analysis showing price versus moving average

Upon selection of an asset, the system executes the trend classification algorithm. This process compares the 50-day Moving Average (short-term trend) against the 200-day Moving Average (long-term trend).

Figure 2 demonstrates this logic visually. The chart displays the price lines, and if the short-term trend line (50MA) crosses above the long-term trend line (200MA), the system classifies the trend as a “BULL” market. Conversely, if it crosses below, it is labeled a “BEAR” market. This provides the user with an immediate, mathematically derived market status.

3. **Set Theory Market Segmentation**

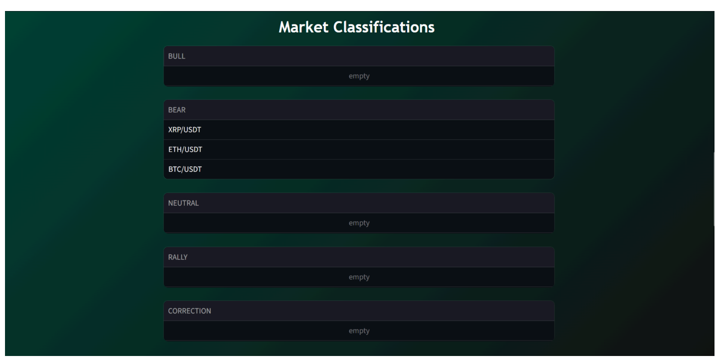

Figure 3. Market Classification displaying assets categorized into sets (Bull, Bear, Neutral, Rally, and Correction) based on trend logic.

The system applies Set Theory to categorize the market landscape. The system iterates through the dictionary of analyzed assets and filters them into distinct sets based on their classification status.

Figure 3 illustrates this segmentation in the user interface. This module displays these sets as distinct groups, allowing users to instantly identify which subset of the market is performing well versus which subset is underperforming.

4. Correlation Graph Visualizer

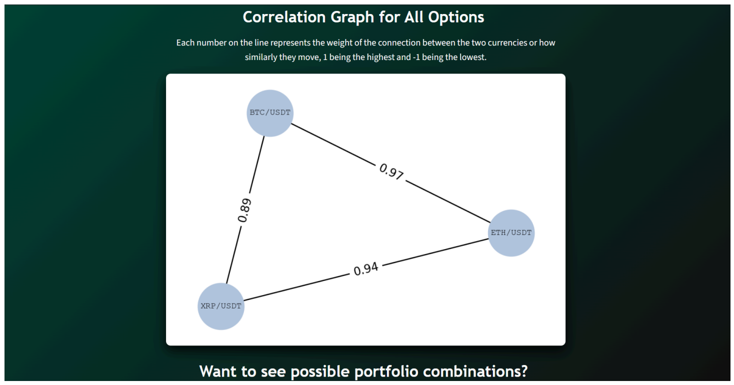

Figure 4. Correlation Graph displaying the correlation strength between assets.

The prototype utilizes the NetworkX library to construct a correlation graph. It identifies whether the different assets move in synchronization.

Figure 4, illustrates this graph structure. The nodes represent individual cryptocurrencies, while the edges represent the strength of the connection between them. High values like 0.97 or 0.89 indicate the price patterns of the two connected currencies are closely identical. This graph assists users in determining if a portfolio is truly correlated to each other.

5. Portfolio Combinatorics and Scoring

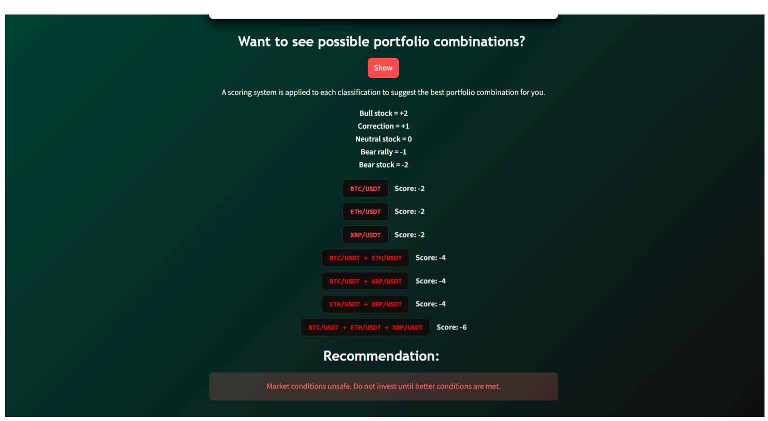

Figure 5. Portfolio combination results and risk assessment warning.

The final module of the prototype applies combinatorics to assist in investment decisions. The program uses itertools library to generate every possible portfolio combination from the selected assets (BTC, ETH, and XRP).

Figure 5 displays the output during unfavorable market conditions. In this figure, the negative scores trigger a warning message. The prototype advises against investment.

CHAPTER 4
RESULTS AND DISCUSSION

Objective 1: Inform users on the possible value trajectory of the cryptocurrency market and what actions to take accordingly.

The program integrated the CCXT library to fetch real-time OHLCV data from the Binance API for analysis and then accurately displayed the data in a way that could be easily understood by users.

Findings: As shown in Figure 2, the program displays the current price of the chosen asset, as well as its increase/decrease percentage in the past 90 days and its classification. Placed below it is a line chart showing the 90-day history of the closing price of the chosen currency, as well as the 50MA and 200MA values across the same time frame. There is also a Streamlit dataframe below the line chart that provides a more detailed insight into its price change rate through the OHLCV data and daily change rate. Lastly, as shown in Figure 3, the program classifies all the options into sets and displays them to the user so they are informed about the current market situation.

Objective 2: Visualize the correlation tendencies and patterns of different cryptocurrency options to help users diversify their portfolio.

The program utilized Graph Theory using the NetworkX Library to calculate and visualize the Correlation Coefficient (r) between assets, helping users identify hidden market connections and optimize their chosen portfolio investments.

Findings: As shown in Figure 4, The results in the visualized graph reveal a positive correlation between the assets that are related to each other during the market cycle on the current day:

- BTC to ETH: 0.97 correlation
- ETH to XRP: 0.94 correlation
- BTC to XRP: 0.89 correlation

A correlation coefficient of 0.97 indicates a near-perfect linear relationship. This finding empirically answers the objective by proving that during this specific timeframe, holding a portfolio of BTC and ETH offered almost no diversification benefit, as both assets moved in synchronization. As for XRP, its price patterns still follow those of BTC closely at 0.89, but its price patterns are closer to ETH at 0.94. All of the results displayed by the graph accurately depict the situation shown by the price line chart above, proving its accuracy.

Objective 3: Suggest possible cryptocurrency investment combinations to boost the portfolio value of users.

The program proved capable in calculating the possible portfolio combinations and suggesting the best course of action to take based on the current market situation, but additional testing is needed.

Findings: Figure 5 shows the combinatorial calculations done by the program. It displays every possible portfolio combination among the available options regardless of number and suggests the best combination for the user to invest. However, during the time this program was being developed, the cryptocurrency market was experiencing a downward trend (all three options experienced a "death cross" in November and had not recovered as of writing). As such, the program accurately displayed negative bear market signals, marking every portfolio combination as negative and suggesting that the user wait for better conditions before investing. This showcases the diagnostic accuracy of the program in Bear classification logic, but the program has not been tested for a positive market and will be unable to do so until the cryptocurrency market experiences an upward trend.

Objective 4: Educate users on market volatility and the potential risks of investing in cryptocurrency to encourage responsible investments among users.

The program fulfilled its objective to educate users on the volatility of the market through showing asset price data in a clear way and providing advice on the best actions to take in the context of the current market.

Findings: Figure 5 shows that the system goes beyond merely showing data and also calculates risk, providing advice based on the data. The program suggests which asset combination is the best to buy at the time, but it also advises the user to hold off on investing in the meantime if the market is experiencing a bearish trend. As shown in the figure, the program detected a bearish trend and displayed a critical message: "Market conditions unsafe. Do not invest until better conditions are met." The warning and combination badges were also colored red to visually emphasize the danger of recklessly investing in any of the affected assets. The line chart in Figure 2 also serves as a visual tool to help the user understand the volatility of the cryptocurrency market and how the program suggests a course of action to the user, especially since it also maps out the 50MA and 200MA lines. As shown in the figure, the chart showed BTC experiencing a "death cross" and dropping to USD84,670 in mid-November, reflecting the asset's poor current state.

Technical Constraints

Regarding the user interface, the development process highlighted certain limitations of Streamlit when modifying the layout with CSS. Since Streamlit uses specific IDs and wrappers for each element, overriding the Streamlit default template's style would require using special commands, inspecting the specific identifiers of each element, or using columns instead of CSS to modify the design for certain elements. As such, the program GUI has certain design limitations, but it is still able to display cryptocurrency data in a user-friendly way due to the features provided by the different imports used in the program.

CHAPTER 5

SUMMARY, CONCLUSION, RECOMMENDATIONS

SUMMARY

The study aimed to address the challenges posed by the highly volatile and unpredictable nature of the cryptocurrency market, which creates obstacles for investors, particularly regarding portfolio diversification and trend analysis. The primary objective was to develop BitPredict, a digital currency insight program that uses discrete mathematical structures specifically Graph Theory, Propositional Logic, Set Theory, and Combinatorics to provide concise market insights. The prototype was built using Python and Streamlit to successfully integrate real-time data from Binance API, classify market trends, and visualize asset correlations. While testing the program, it revealed a high correlation between all three assets, ranging from 0.89 to 0.97, and accurately detected a “death cross” or bear market, leading the system to correctly recommend users against investing thereby fulfilling the objectives of the study.

CONCLUSION

This study was able to successfully achieve the objective of developing BitPredict, a prototype that utilizes discrete mathematical structures to provide transparent insights into the unpredictable market of cryptocurrency. By integrating Graph Theory, Propositional Logic, Set Theory, and Combinatorics, the program effectively served as a bridge between complex predictions and accessible financial analysis.

The results of the tests further validated the system's accuracy and helpfulness in real-world scenarios. To be exact, the addition of Graph Theory revealed a significant correlation coefficient between all of the sample cryptocurrencies, demonstrating the lack of diversification benefits between them during the specific observation period. Furthermore, the trend classification algorithms, which utilized propositional logic, identified a bear market trend characterized by a "death cross", which was able to trigger warning mechanisms that would advise the users to stop investments until improvements are seen.

In conclusion, BitPredict was able to show that financial analysis tools do not need to rely on deep learning models. Instead, discrete structures can help clarify market behaviors. The prototype was able to fulfill its intended purpose by being able to translate raw financial data into risk assessments. With all of these combined, users will be able to navigate the unstable market of cryptocurrency with greater awareness and a sense of financial security.

RECOMMENDATION

Based on the findings after conducting tests on the developed program, here are the following recommendations for improvements:

1. The program has only been tested on a small dataset of cryptocurrencies that have roughly the same movement. If used on larger datasets, the simple set scoring system might need revisions since it may prove ineffective and require a significant amount of space and time in calculating the best possible investment option. The graph may also need revisions to properly accommodate larger datasets of differing movement trends.
2. The program has, as of writing, only been tested for bear market runs due to the current cryptocurrency market situation and the program tracking real-time data. One possible way to circumvent this, instead of waiting for the market to recover, could be to use a backtesting module to simulate how the program would process data from past time periods where the market was experiencing an upward trend.
3. Streamlit offers limited GUI customization options due to certain enforced rigidities within Streamlit itself. If improvements to the interface are to be made, it would be recommended to move to a more flexible framework that offers more customizability and freedom.
4. The 200MA and 50MA comparison method is a reliable method for detecting signals, but it uses past data to classify trends, so there is a risk that the program's declaration of a market trend may be outdated. To minimize the risk of delayed reactions to volatile market trends, the program's propositional logic could incorporate other leading indicators like the Relative Strength Index (RSI) and the Moving Average Convergence/Divergence (MACD) in later versions.

CITED REFERENCES

Indhumathi V., Kumar R. S. (2023). Crypto-Currency App Using React. International Journal of Research Publication and Reviews. 2023;4(6):3294–3299. https://ijrpr.com/uploads/V4ISSUE6/IJRPR14522.pdf

Ramesh, N., Aishwarya, V. (2025). Cryptocurrency Awareness and Investment Behaviour among Millennial Investors. 10.13140/RG.2.2.12331.50722. https://www.researchgate.net/publication/395672580_Cryptocurrency_Awareness_and_Investment_Behaviour_among_Millennial_Investors

Shewarade, Y., Aly, S., Yoon, H. J., & Chung, S. (2025). CryptoProphet: Building a Cryptocurrency Portfolio App with Integrated Market Predictive Models. *Journal of Information Systems Applied Research and Analytics*, *18*(3), 4–17. https://doi.org/10.62273/nnld2781

Reiff N. Bitcoin's price fell $30K+ since last Month—Why is it so volatile? Investopedia. 2025 Nov 18. https://www.investopedia.com/bitcoins-price-since-last-month-why-is-it-so-volatile-11850923

What to know about cryptocurrency and scams. Federal Trade Commission. 2025 Feb 10. https://consumer.ftc.gov/articles/what-know-about-cryptocurrency-scams

Hatze V. How many cryptocurrencies are there? Number of crypto tokens in 2025. Demo. 2025 Sep 11. https://yieldfund.com/how-many-cryptocurrencies-are-there-number-of-crypto-tokens-in-2025/

Nallapaneni D. Why does all crypto move together? | CoinLedger. CoinLedger. 2025 Oct 27. https://coinledger.io/learn/why-does-all-crypto-move-together

Lower BTC vs ETH correlation. Coinbase. 2023 Apr 21. https://www.coinbase.com/institutional/research-insights/research/weekly-market-commentary/lower-btc-vs-eth-correlation-april-21-2023

Nguyen, A. P. N., Mai, T. T., Bezbradica, M., & Crane, M. (2023). Volatility and returns connectedness in cryptocurrency markets: Insights from graph-based methods. Physica A: Statistical Mechanics and Its Applications, 632(1), 129349. https://doi.org/10.1016/j.physa.2023.129349

Puerto, J., Ricca, F., Rodríguez-Madrena, M., & Scozzari, A. (2021). A combinatorial optimization approach to scenario filtering in portfolio selection. arXiv.org. https://doi.org/10.48550/arXiv.2103.01123

Karl, T. (2024, July 30). Benefits of Python for data analytics explained. *New Horizons*. https://www.newhorizons.com/resources/blog/benefits-of-python-for-data-analytics

Maverick, J. B. (2025, October 10). *Understanding the 200-Day SMA: Key indicator for market trends*. Investopedia. https://www.investopedia.com/ask/answers/013015/why-200-simple-moving-average-sma-so-common-traders-and-analysts.asp

Nickolas, S. (2024, December 26). *Correlation coefficients: positive, negative, and zero*. Investopedia. https://www.investopedia.com/ask/answers/032515/what-does-it-mean-if-correlation-coefficient-positive-negative-or-zero.asp

Indeed Editorial Team. (2025, July 24). *What Is Backtesting? Plus Pros and Cons and Examples*. Indeed. https://www.indeed.com/career-advice/career-development/backtesting

Hayes, A. (2025, April 24). *Golden Cross pattern explained with examples and charts*. Investopedia. https://www.investopedia.com/terms/g/goldencross.asp

Golden Cross Trading Strategy: A complete guide | Capital.com UK. (n.d.). Capital.com. https://capital.com/en-gb/learn/trading-strategies/golden-cross-trading-strategy

Maverick, J. (2025, November 21). *MACD vs. RSI: Key Differences and Uses for Traders*. Investopedia. https://www.investopedia.com/ask/answers/122214/what-are-main-differences-between-moving-average-convergence-divergence-macd-relative-strength-index.asp

APPENDICES

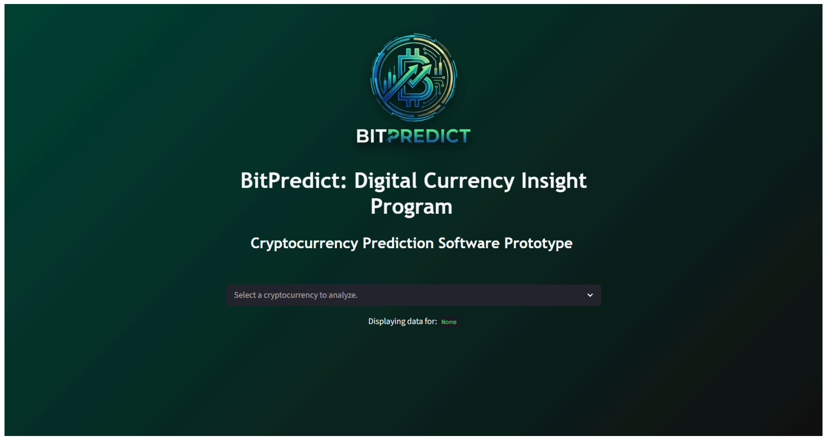

Figure 1. BitPredict Landing Page and User Interface Entry Point.

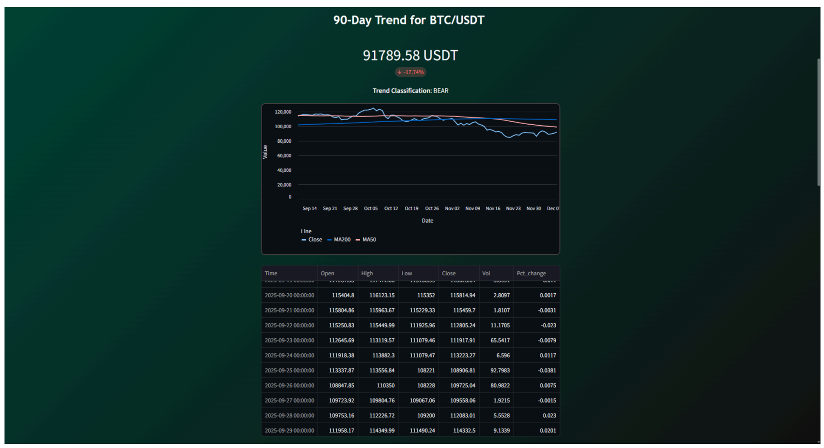

Figure 2. Single cryptocurrency asset analysis showing price versus moving average

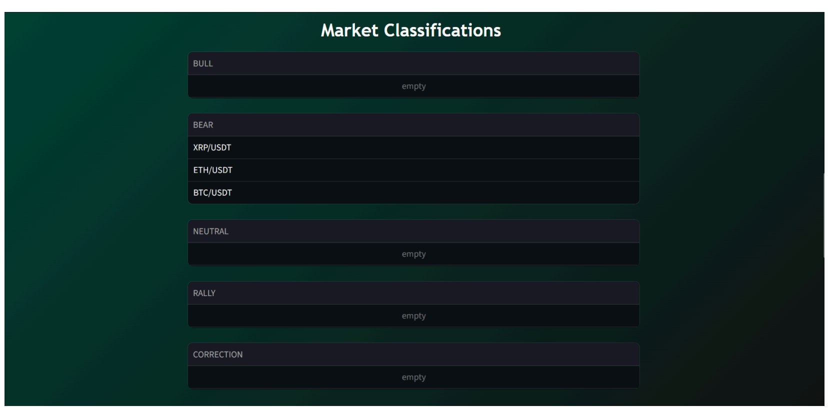

Figure 3. Market Classification displaying assets categorized into sets (Bull, Bear, Neutral, Rally, and Correction) based on trend logic.

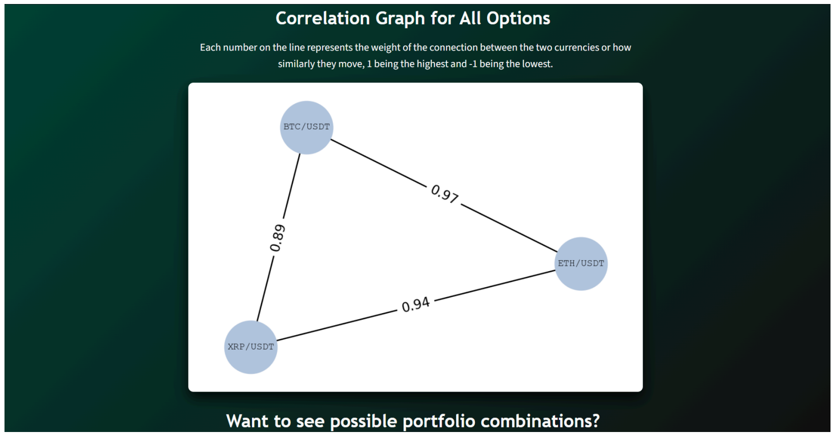

Figure 4. Correlation Graph displaying the correlation strength between assets.

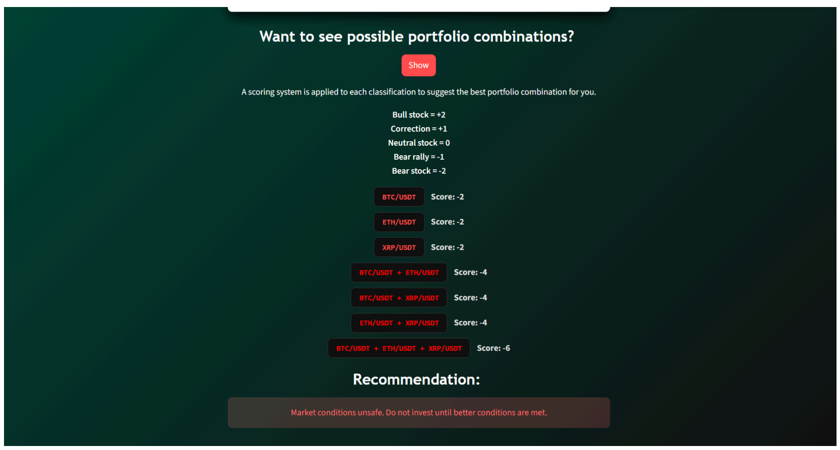

Figure 5. Portfolio combination results and risk assessment warning.